JNTUH SEP 13-06-18AM Set 2

R13

Code No: 118BA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2018
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) What are fraudulent threats? [2]
b) What are the access modes in Acten model? Briefly Explain. [3]
c) What is the basic property of Smith security model? [2]
d) Write about implicit flow in Lattice model. [3]
e) List out various types of attacks on database. [2]
f) Define object granularity. Explain how it is varied in OS and DBMS. [3]
g) What are the entities of SORION model? [2]
h) What is Role lattice? [3]
i) Write the basic principle of message filter. [2]
j) List the issue of authorization in active databases. [3]

PART - B

(50 Marks)

2.a) What is Access control? Write the role of access control in data base security.
b) Discuss about Mandatory and Discretionary access controls. [5+5]

OR

3.a) Explain about inference control.
b) Explain Authorization states, access modes and operations in Take Grant model. [5+5]

4.a) What are system states and operations of Bell and Lapadu's security model?
b) Explain about access control mechanisms in data base. [5+5]

OR

5.a) Give detailed note on MAC secure model.
b) Explain register based memory protection. [5+5]

6.a) Explain kernlized architecture with a neat diagram.
b) Give a note on restriction based techniques of inference protection. [5+5]

OR

7.a) What are the different security mechanism in DBMS? Explain each mechanism briefly.
b) Explain IDES model with a neat diagram. [5+5]

8.a) Explain the elements of object oriented database.
b) What is frame based system? Explain, how the security is provided to frame based system? [5+5]

OR

9.a) Explain security policy axioms of SORION model.
b) With the examples, explain various constraints supported by SORION. [5+5]

10.a) Illustrate how ORION model considers characteristics of object oriented system such as inheritance, composite object and hierarchy.
b) What is message filtering? Explain the entities of message filtering. [5+5]

OR

11.a) Write the elements of active data bases.
b) Write the implication rules for deriving the implicit authorization in ORION model.[5+5]

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